



MECHATRONICS
SPECTRUM
TEAM



MAE

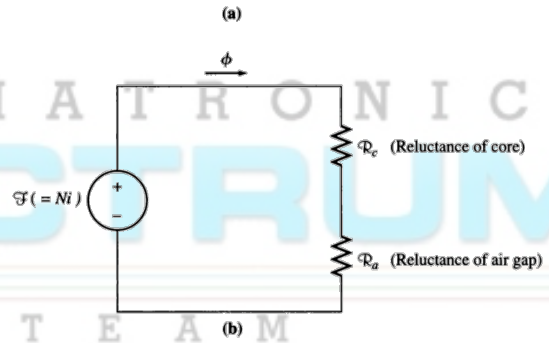
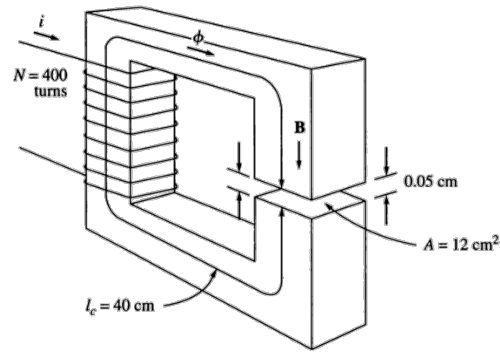
ELECTRICAL MACHINES I RULES SUMMARY

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SPECTRUM TEAM**

CHAPTER 1:

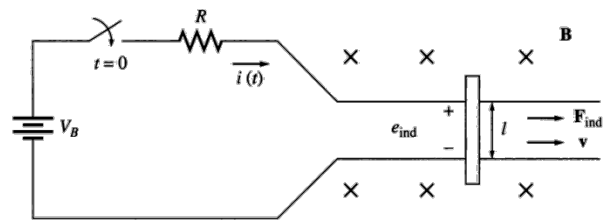
• Metal Core Rules:

- $H = \frac{Ni}{l_c}$ (A·turn/m)
- $B = \mu H$ (T)
- $\mu = \mu_0 \mu_r$ (H/m)
- $\phi = BA$ (Wb)
- $\mathcal{F} = Ni$ (A·turn)
- $\mathcal{F} = \phi \mathcal{R}$ (A·turn)
- $\mathcal{R} = \frac{l_c}{\mu A}$ (A·turn/Wb)



• Induces Voltage and force:

- $e_{ind} = -N \frac{d\phi}{dt}$ (V)
- $e_{ind} = v l B \sin \theta$ (V)
- $F_{ind} = i l B \cos \theta$ (N)



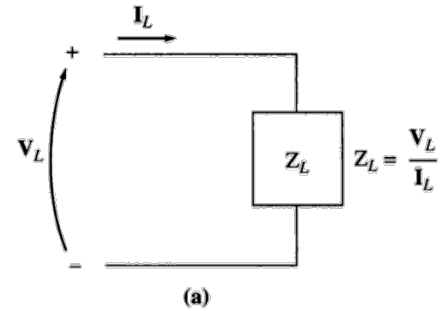
• Linear DC Machine:

- $F_{app} = F_{ind} = i l B$
- $i_{max} = \frac{V_B}{R}$
- $e_{ind} = V_B + iR$ if $(F: \rightarrow, B: \times \mid F: \leftarrow, B: \cdot)$ → Generator:
- $e_{ind} = V_B - iR$ if $(F: \leftarrow, B: \times \mid F: \rightarrow, B: \cdot)$ → Motor:
- $v_{ss} = \frac{e_{ind}}{lB}$
- At no load → $(e_{ind} = V_B, v_{ss} = \frac{V_B}{lB}, F_{ind} = 0, i = 0)$

CHAPTER 2:

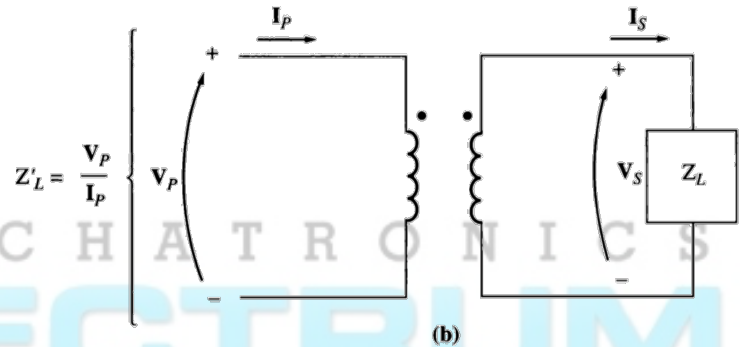
- **Basics:**

- $S = VI$ (VA)
- $P = VI \cos \theta = I^2 R$ (W)
- $Q = VI \sin \theta = I^2 X$ (VA)
- $PF = \cos \theta$



- **Ideal Transformer:**

- $\frac{v_p}{v_s} = \frac{N_p}{N_s} = \frac{i_s}{i_p} = a$
- $Z' = a^2 Z$ "Referred to primary"

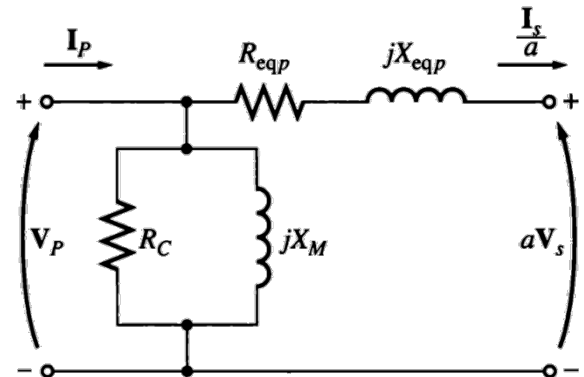


- **Short-Circuit Test:**

- $\theta_{SC} = \cos^{-1} \left(\frac{P_{SC}}{V_{SC} \cdot I_{SC}} \right)$
- $Z_{SE} = \frac{V_{SC}}{I_{SC}} \angle \theta_{SC} \rightarrow R_{SE} + jX_{SE}$

- **Open-Circuit Test:**

- $\theta_{OC} = \cos^{-1} \left(\frac{P_{OC}}{V_{OC} \cdot I_{OC}} \right)$
- $Y_E = \frac{I_{OC}}{V_{OC}} \angle -\theta_{OC} \rightarrow \frac{1}{R_C} - j \frac{1}{X_M}$



- **Per Unit:**

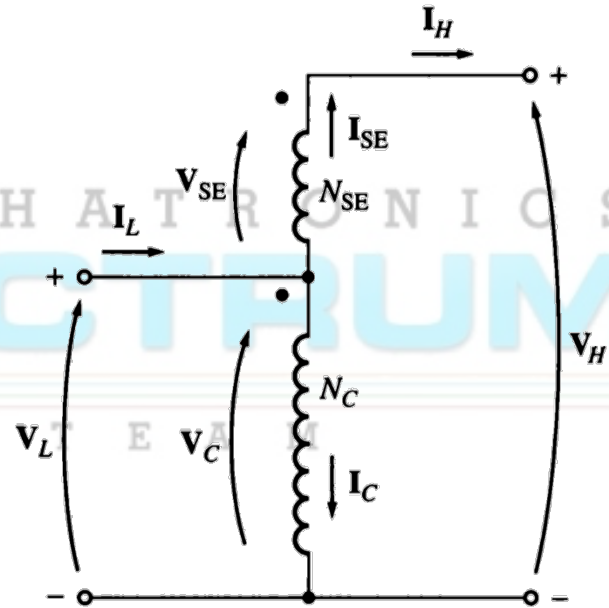
- $x_{PU} = \frac{x_{Actual}}{x_{Base}}$ (unitless)

- **Voltage Regulation:**

- $VR = \frac{V_{nl} - V_{fl}}{V_{fl}} \cdot 100\%$
- $I_s = \frac{S}{V_s}, \theta = \angle \cos^{-1} \text{PF}, \theta \rightarrow \begin{cases} +, \text{ leading PF} \\ -, \text{ lagging PF} \end{cases}$
- $V_{nl} = V_{fl} + (I_s \angle \theta)(R_{SE} + jX_{SE})$ "Referred to secondary"
- $V_p = a V_{nl}$

- **Auto Transformer:**

- $\frac{V_L}{V_H} = \frac{I_H}{I_L} = \frac{N_C}{N_C + N_{SE}}$
- $S_{in} = V_H I_H = V_L I_L = S_{out}$
- $S_W = V_C I_C = V_{SE} I_{SE}$
- $\frac{S_{io}}{S_W} = \frac{N_C + N_{SE}}{N_{SE}}$



- **3-Phase Transformer:**

- $Y - Y \rightarrow \frac{V_{Lp}}{V_{Ls}} = \frac{\sqrt{3} V_{\phi p}}{\sqrt{3} V_{\phi s}} = a$
- $Y - \Delta \rightarrow \frac{V_{Lp}}{V_{Ls}} = \frac{\sqrt{3} V_{\phi p}}{V_{\phi s}} = \sqrt{3} a$
- $\Delta - Y \rightarrow \frac{V_{Lp}}{V_{Ls}} = \frac{V_{\phi p}}{\sqrt{3} V_{\phi s}} = \frac{a}{\sqrt{3}}$
- $\Delta - \Delta \rightarrow \frac{V_{Lp}}{V_{Ls}} = \frac{V_{\phi p}}{V_{\phi s}} = a$

CHAPTER 7:

• Simple DC Machine:

- $\tau_{ind} = (2/\pi)\phi i = (2rlB) i$
- $e_{ind} = (2/\pi)\phi\omega = (2rlB) \omega$
- $i = \frac{\tau}{2rlB}$
- $e_{ind} = V_B + iR$ at $(\tau + \text{unloaded}) \rightarrow \text{Generator}$
- $e_{ind} = V_B - iR$ at $(\tau + \text{loaded}) \rightarrow \text{Motor}$
- $\omega_{ss} = \frac{e_{ind}}{2rlB}$ (rad/s)

• Rotor and commutator:

$Z \rightarrow$ #conductors, $C \rightarrow$ #coils, $N_C \rightarrow$ #turns per coil
 $p \rightarrow$ fractional pitch, $P \rightarrow$ #poles, $a \rightarrow$ #current pathes
 $y_c \rightarrow$ commutator pitch, $m \rightarrow$ #plex.
" $\pm$ " $\rightarrow$ $\begin{cases} + \rightarrow \text{progressive} \\ - \rightarrow \text{retrogressive} \end{cases}$

- $Z = 2CN_C$, $p = \frac{\theta_e}{180^\circ}$

• Lap Winding:

- $y_c = \pm m$
- $a = mP$

• Wave Winding:

- $y_c = \frac{2(C \pm 1)}{P}$
- $a = 2m$

• Armature Reaction:

- $K = \frac{ZP}{2\pi a} @ \omega_m$ (rad/s)
- $K' = \frac{ZP}{60a} @ n_m$ (RPM)
- $E_A = K\phi\omega_m = K'\phi n_m$
- $\tau_{ind} = K\phi I_A$

• Power Losses:

- $\eta = \frac{P_{in} - P_{loss}}{P_{in}} \cdot 100\%$
- $P_{A loss} = I_A^2 R_A$, $P_{F loss} = I_F^2 R_F$
- $P_{conv} = \tau_{ind} \omega_m = E_A I_A$
- $SR = \frac{\omega_{nl} - \omega_{fl}}{\omega_{fl}} \cdot 100\%$

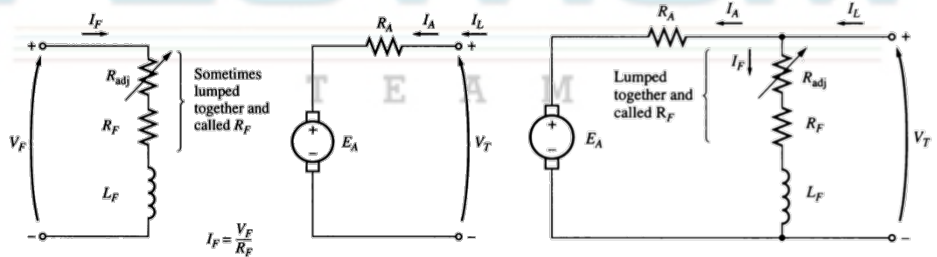
CHAPTER 8:

• Separately Excited & Shunt DC Motors:

$$I_F = \begin{cases} \frac{V_T}{R_F} \rightarrow \text{Shunt} \\ \frac{V_F}{R_F} \rightarrow \text{Separately Excited} \end{cases}, \quad I_F^* = I_F - \frac{\mathcal{F}_A}{N_F}$$

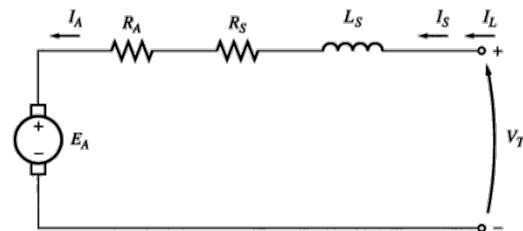
“ E_{A0} from characteristic curve when there are $\mathcal{F}_A$ ”

- $E_A = V_T - I_A R_A$
- $I_A = I_L - I_F$
- $\frac{\omega_m}{\omega_{m0}} = \frac{n_m}{n_{m0}} = \frac{E_A}{E_{A0}}$
- $\tau_{ind} = \frac{E_A I_A}{\omega_m}$



• Series DC Motor:

- $E_A = V_T - I_A (R_A + R_S)$
- $I_A = I_S = I_L$
- $\mathcal{F}_A = N I_A$



“ E_{A0} from characteristic curve at $\mathcal{F}_A$ ”

- $\frac{\omega_m}{\omega_{m0}} = \frac{n_m}{n_{m0}} = \frac{E_A}{E_{A0}}$
- $\tau_{ind} = \frac{E_A I_A}{\omega_m} = K_c I_A^2$
- $\omega_m = \frac{V_T}{\sqrt{K_c \tau_{ind}}} - \frac{R_A + R_S}{K_c}$

• Separately Excited & Shunt DC Generator:

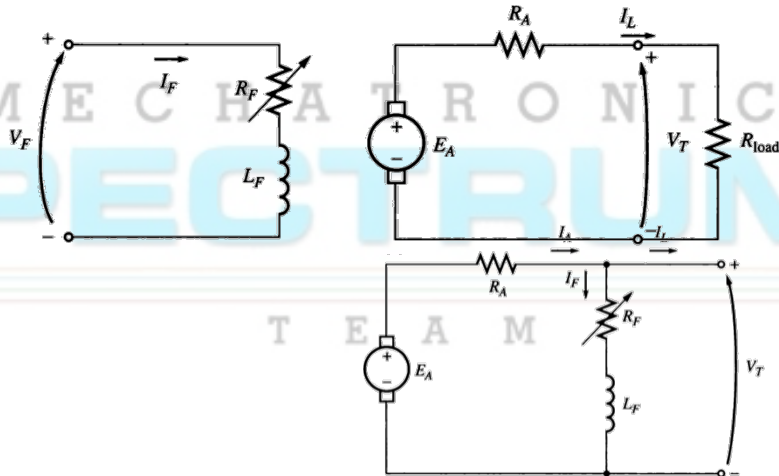
$$I_A = \begin{cases} I_F + I_L = \frac{V_T}{R_F} + I_L \rightarrow \text{Shunt} \\ I_F = \frac{V_F}{R_F} \rightarrow \text{Separately Excited} \end{cases}, \quad I_F^* = I_F - \frac{\mathcal{F}_A}{N_F}$$

“ E_{A0} from characteristic curve when there are $\mathcal{F}_A$ ”

$$- E_A = V_T + I_A(L^*)R_A$$

$$- \frac{\omega_m}{\omega_{m0}} = \frac{n_m}{n_{m0}} = \frac{E_A}{E_{A0}}$$

$$- \tau_{ind} = \frac{E_A I_A}{\omega_m}$$

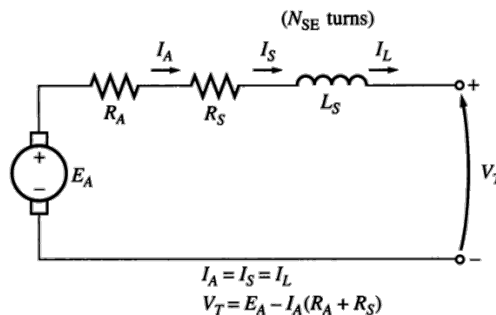


• Series DC Generator:

$$- E_A = V_T + I_A(R_A + R_S)$$

$$- I_A = I_S = I_L$$

$$- \mathcal{F}_A = N I_A$$



“ E_{A0} from characteristic curve at $\mathcal{F}_A$ ”

$$- \frac{\omega_m}{\omega_{m0}} = \frac{n_m}{n_{m0}} = \frac{E_A}{E_{A0}}$$

$$- \tau_{ind} = \frac{E_A I_A}{\omega_m} = K_c I_A^2$$

$$- \omega_m = \frac{V_T}{\sqrt{K_c \tau_{ind}}} - \frac{R_A - R_S}{K_c}$$

Green: Rules Differences Between Motor & Generator.